

Lecture 19: Normal Matrices

Hermitian Geometry and Spectral Projections

Today's Question

From previous lectures, spectral decomposition tries to split a matrix into pieces:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_k P_k.$$

Then every vector splits into eigenspace components:

$$\mathbf{v} = P_1 \mathbf{v} + P_2 \mathbf{v} + \cdots + P_k \mathbf{v}.$$

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Are these pieces perpendicular?

Or are they only oblique algebraic pieces?

Why Study Normal Matrices?

Diagonalizable matrices already give spectral decomposition.

But diagonalization alone does **not** promise geometry:

$$\mathbb{C}^n = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k} \quad \text{may be oblique.}$$

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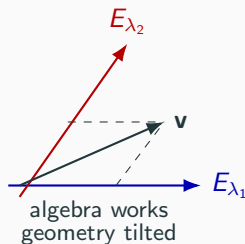
Normal matrices are the matrices where spectral decomposition becomes **orthogonal geometry**:

$$\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} E_{\lambda_2} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.$$

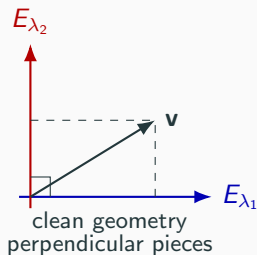
This is why normal matrices matter.

Two Pictures to Keep in Mind

Oblique eigenspaces



Orthogonal eigenspaces



Normal matrices answer: when does the right picture happen?

The Surprising Theorem

Main Theorem We Will Prove

For a complex matrix A ,

A normal $\iff$ all spectral projections P_i satisfy $P_i^H = P_i$.

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Equivalently:

$$A \text{ normal} \iff \text{eigenspaces of } A \text{ are mutually Hermitian-orthogonal.}$$

Not just diagonalizable: diagonalizable with orthogonal spectral pieces.

How This Connects to Matrix Exponentials

Lecture 18 gave the spectral formula:

$$e^{As} = e^{\lambda_1 s} P_1 + \cdots + e^{\lambda_k s} P_k.$$

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Lecture 18 gave the spectral formula:

$$e^{As} = e^{\lambda_1 s} P_1 + \cdots + e^{\lambda_k s} P_k.$$

If the projections are oblique, the algebra works but the geometry is distorted.

If A is normal, then each P_i is an orthogonal projection:

$$P_i^H = P_i.$$

Meaning: the system evolves along perpendicular eigenspace directions.

Complex Length

Why Are We in the Complex World?

Our spectral method needs roots of the characteristic polynomial:

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Over $\mathbb{C}$, every polynomial factors into linear factors.

So the spectral story naturally lives over $\mathbb{C}$.

But Complex Geometry Needs Repair

We want to measure:

- **length**: how big is a vector?
- **angle**: when are two vectors perpendicular?

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Over $\mathbb{C}$, this formula breaks length.

Hermitian geometry is the repaired geometry of complex vector spaces.

The Old Length Formula

Over real numbers, length comes from

$$\|\mathbf{v}\|^2 = \mathbf{v}^T \mathbf{v}.$$

For

$$\mathbf{v} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n,$$

we get

$$\mathbf{v}^T \mathbf{v} = x_1^2 + x_2^2 + \cdots + x_n^2.$$

Key: a nonzero real vector has positive length squared.

What Goes Wrong over $\mathbb{C}$?

Try the same formula on a complex vector:

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

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Naive length squared:

$$\mathbf{v}^T \mathbf{v} = \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = 1 + i^2.$$

But $i^2 = -1$, so

$$\mathbf{v}^T \mathbf{v} = 1 - 1 = 0.$$

Problem: a nonzero vector has zero length.

Start in One Complex Dimension

For one complex number $z = a + bi$, ordinary absolute value satisfies

$$|z|^2 = a^2 + b^2.$$

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The algebraic way to get this number is

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So complex length already contains the rule:

$$|z|^2 = \bar{z}z.$$

Hermitian length is the vector version of this one-dimensional fact.

The Missing Move

The problem is this multiplication:

$$i \cdot i = i^2 = -1.$$

For length, we want a positive number:

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So before taking the dot product, we must mirror the complex number:

$$i \longmapsto -i.$$

Then

$$(-i) \cdot i = 1.$$

Key: complex length needs conjugation before dot product.

Hermitian Transpose of a Vector

Definition: Hermitian Transpose of a Vector

For a complex column vector $\mathbf{v}$, define

$$\mathbf{v}^H := \overline{\mathbf{v}}^T.$$

Immediate example:

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix} \implies \mathbf{v}^H = \begin{pmatrix} 1 & -i \end{pmatrix}.$$

The letter H means **Hermitian**.

The Same Vector Now Has Positive Length

For

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \mathbf{v}^H = \begin{pmatrix} 1 & -i \end{pmatrix}.$$

Now compute:

$$\mathbf{v}^H \mathbf{v} = \begin{pmatrix} 1 & -i \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = 1 + (-i)i = 2.$$

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No zero-length nonzero vector.

Key: H repairs geometry over $\mathbb{C}$.

Hermitian Inner Product

Definition: Hermitian Inner Product

For complex vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$, define

$$\langle \mathbf{v}, \mathbf{w} \rangle_H := \mathbf{v}^H \mathbf{w}.$$

If

$$\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix},$$

then

$$\langle \mathbf{v}, \mathbf{v} \rangle_H = \mathbf{v}^H \mathbf{v} = |v_1|^2 + \cdots + |v_n|^2.$$

Key: this is always positive when $\mathbf{v} \neq \mathbf{0}$.

Hermitian Orthogonal Vectors

Definition: Hermitian Orthogonal

Two complex vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ are **Hermitian orthogonal** if

$$\mathbf{v}^H \mathbf{w} = 0.$$

Example:

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

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Hermitian Orthogonal Subspaces

Definition: Hermitian Orthogonal Subspaces

Two subspaces $V, W \subseteq \mathbb{C}^n$ are **Hermitian orthogonal** if

$$\mathbf{v}^H \mathbf{w} = 0$$

for every $\mathbf{v} \in V$ and every $\mathbf{w} \in W$.

Notation:

$$V \perp_H W.$$

This is the geometry we want for eigenspaces.

Hermitian Conjugate

Hermitian Conjugate of a Matrix

Definition: Hermitian Conjugate

For a complex matrix A , define

$$A^H := \overline{A^T}.$$

This means:

transpose first, then **conjugate every entry**.

This is also called the **conjugate transpose**.

Concrete Example

Let

$$A = \begin{pmatrix} 1 & i \\ 2-i & 3 \end{pmatrix}.$$

First transpose:

$$A^T = \begin{pmatrix} 1 & 2-i \\ i & 3 \end{pmatrix}.$$

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First transpose:

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Then conjugate:

$$A^H = \overline{A^T} = \begin{pmatrix} 1 & 2+i \\ -i & 3 \end{pmatrix}.$$

Real Matrices Are the Old Case

If all entries of A are real, conjugation changes nothing.

So

$$A^H = \overline{A^T} = A^T.$$

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Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \implies \quad A^H = A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

Key: Hermitian conjugate is the complex version of transpose.

The Product Rule Reverses Order

For matrices,

$$(AB)^H = B^H A^H.$$

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Why order reverses:

$$(AB)^H = \overline{(AB)}^T = \overline{B^T A^T} = \overline{B^T} \overline{A^T} = B^H A^H.$$

Key: Hermitian conjugate reverses multiplication order.

Why A^H Is the Matrix-Level Mirror

We stay at the matrix level.

The only operation we need is:

$$(A\mathbf{v})^H \mathbf{w} = \mathbf{v}^H A^H \mathbf{w}.$$

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We stay at the matrix level.

The only operation we need is:

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In inner-product notation this says:

$$\langle A\mathbf{v}, \mathbf{w} \rangle_H = \langle \mathbf{v}, A^H \mathbf{w} \rangle_H.$$

No extra abstract structure is being introduced: A^H is just conjugate transpose.

Normal Matrices

Normal Matrices

Definition: Normal Matrix

A complex square matrix A is **normal** if

$$AA^H = A^H A.$$

In words:

A commutes with its Hermitian conjugate.

If A is real, then $A^H = A^T$, so normal means

$$AA^T = A^T A.$$

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This is not the final theorem.

The real question: what does this condition do to spectral projections?

First Special Case: Hermitian Symmetric

Definition: Hermitian Symmetric

A matrix A is **Hermitian symmetric** if

$$A^H = A.$$

Example:

$$A = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix}, \quad A^H = A.$$

If all entries are real, this is exactly a real symmetric matrix: $A^T = A$.

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If all entries are real, this is exactly a real symmetric matrix: $A^T = A$.

Then A is automatically normal:

$$AA^H = A^2 = A^H A.$$

Second Special Case: Skew-Hermitian Symmetric

Definition: Skew-Hermitian Symmetric

A matrix A is **skew-Hermitian symmetric** if

$$A^H = -A.$$

Example:

$$A = \begin{pmatrix} i & 2+i \\ -2+i & 3i \end{pmatrix}, \quad A^H = -A.$$

If all entries are real, this is exactly a real skew-symmetric matrix: $A^T = -A$.

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If all entries are real, this is exactly a real skew-symmetric matrix: $A^T = -A$.

Then A is automatically normal:

$$AA^H = A(-A) = -A^2 = (-A)A = A^HA.$$

Third Special Case: Unitary

Definition: Unitary Matrix

A matrix U is **unitary** if

$$U^H U = U U^H = I.$$

Example:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

If all entries are real, this is exactly a real orthogonal matrix:

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If all entries are real, this is exactly a real orthogonal matrix:

$$U^T U = U U^T = I.$$

Unitary matrices are automatically normal:

$$U U^H = I = U^H U.$$

What Does $U^H U = I$ Mean?

Write the columns of U as

$$U = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \\ | & | & \cdots & | \end{pmatrix}.$$

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Write the columns of U as

$$U = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \\ | & | & \cdots & | \end{pmatrix}.$$

Then $U^H U$ is the Hermitian inner-product table of the columns:

$$U^H U = \begin{pmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle_H & \langle \mathbf{u}_1, \mathbf{u}_2 \rangle_H & \cdots \\ \langle \mathbf{u}_2, \mathbf{u}_1 \rangle_H & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle_H & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

So $U^H U = I$ means the columns form a Hermitian orthonormal basis.

A Matrix That Is Not Normal

Let

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since J is real,

$$J^H = J^T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

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$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since J is real,

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Compute both orders:

$$JJ^H = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad J^H J = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

So $JJ^H \neq J^H J$. This matrix is **not normal**.

Diagonalizability

Goal Before Spectral Projections

We want to use spectral projections:

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$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

But spectral projections are legitimate only after we know:

A is diagonalizable.

So the first goal is:

Theorem

Every normal matrix over $\mathbb{C}$ is diagonalizable.

Recall the Practical Criterion

Write the characteristic polynomial honestly as a determinant:

$$\det(tI - A) = \prod_{i=1}^k (t - \lambda_i)^{m_i}.$$

Remove repeated powers:

$$r_A(t) = \prod_{i=1}^k (t - \lambda_i).$$

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Previous lecture criterion:

$$A \text{ diagonalizable} \iff r_A(A) = 0.$$

Trace of the Goal

The goal is not a list of procedures.

The goal is one obstruction:

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The goal is one obstruction:

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So for a normal matrix, everything concentrates on this expression:

$$\boxed{r_A(A)}.$$

Question: what kind of matrix is produced by plugging A into this radical polynomial?

Nilpotent Matrices

Definition: Nilpotent Matrix

A matrix N is **nilpotent** if some power of it is zero:

$$N^m = 0$$

for some positive integer m .

Example:

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

Nilpotent means: after enough repeated multiplication, the matrix disappears.

What Always Happens to $r_A(A)$?

Write

$$\det(tI - A) = \prod_{i=1}^k (t - \lambda_i)^{m_i}, \quad r_A(t) = \prod_{i=1}^k (t - \lambda_i).$$

What Always Happens to $r_A(A)$?

Write

$$\det(tI - A) = \prod_{i=1}^k (t - \lambda_i)^{m_i}, \quad r_A(t) = \prod_{i=1}^k (t - \lambda_i).$$

Let

$$M = \max\{m_1, \dots, m_k\}.$$

Then every exponent in $r_A(t)^M$ is large enough:

$$r_A(t)^M = \prod_{i=1}^k (t - \lambda_i)^M = \det(tI - A) q(t).$$

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Then every exponent in $r_A(t)^M$ is large enough:

$$r_A(t)^M = \prod_{i=1}^k (t - \lambda_i)^M = \det(tI - A) q(t).$$

Plug in A :

$$r_A(A)^M = \det(tI - A)|_{t=A} q(A) = 0.$$

So $r_A(A)$ is always nilpotent.

The Strategy Trace

For every matrix A :

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For every matrix A :

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For a normal matrix A , we want one more property:

$r_A(A)$ is normal.

Then the whole theorem reduces to one collapse:

$$\boxed{\text{normal} + \text{nilpotent} \implies 0.}$$

After that collapse,

$$r_A(A) = 0,$$

so A is diagonalizable.

Normal Nilpotent Matrices Must Vanish

Assume N is normal and nilpotent.

Nilpotent gives some power zero. Choose a power of two beyond it:

$$N^{2^r} = 0.$$

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Nilpotent gives some power zero. Choose a power of two beyond it:

$$N^{2^r} = 0.$$

Because N is normal,

$$NN^H = N^H N,$$

so powers of N commute with powers of N^H .

We will descend from $N^{2^r} = 0$ to $N = 0$.

The Descent Calculation

Start with

$$N^{2^r} = 0.$$

Multiply by $(N^H)^{2^r}$:

$$N^{2^r} (N^H)^{2^r} = 0.$$

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Multiply by $(N^H)^{2^r}$:

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Since powers commute, fold the product as

$$(N^{2^{r-1}} (N^H)^{2^{r-1}}) (N^{2^{r-1}} (N^H)^{2^{r-1}}) = 0.$$

Set

$$M = N^{2^{r-1}}.$$

Then the equation becomes

$$(MM^H)(MM^H)^H = 0.$$

The Hidden Tool Appears

The proof now needs one simple fact:

$$XX^H = 0 \implies X = 0.$$

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Why? Write the rows of X as colored vectors:

$$X = \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \\ \vdots \end{pmatrix}.$$

Then XX^H is the Hermitian inner-product table of these row vectors.

The Inner-Product Table

$$XX^H = \begin{pmatrix} \langle \mathbf{r}_1, \mathbf{r}_1 \rangle_H & \langle \mathbf{r}_1, \mathbf{r}_2 \rangle_H & \langle \mathbf{r}_1, \mathbf{r}_3 \rangle_H & \cdots \\ \langle \mathbf{r}_2, \mathbf{r}_1 \rangle_H & \langle \mathbf{r}_2, \mathbf{r}_2 \rangle_H & \langle \mathbf{r}_2, \mathbf{r}_3 \rangle_H & \cdots \\ \langle \mathbf{r}_3, \mathbf{r}_1 \rangle_H & \langle \mathbf{r}_3, \mathbf{r}_2 \rangle_H & \langle \mathbf{r}_3, \mathbf{r}_3 \rangle_H & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

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If $XX^H = 0$, then the diagonal entries are zero:

$$\|\mathbf{r}_1\|_H^2 = 0, \quad \|\mathbf{r}_2\|_H^2 = 0, \quad \|\mathbf{r}_3\|_H^2 = 0, \quad \dots$$

So every row vector is zero, hence $X = 0$.

Use the Tool Twice

We had

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Apply the table fact to

$$X = MM^H.$$

This gives

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$$(MM^H)(MM^H)^H = 0.$$

Apply the table fact to

$$X = MM^H.$$

This gives

$$MM^H = 0.$$

Apply the same table fact again to

$$X = M.$$

This gives

$$M = 0.$$

Since $M = N^{2^{r-1}}$,

$$N^{2^r} = 0 \implies N^{2^{r-1}} = 0.$$

Repeat the Descent

The same argument removes one half of the exponent each time:

$$N^{2^r} = 0 \implies N^{2^{r-1}} = 0 \implies N^{2^{r-2}} = 0 \implies \dots \implies N^2 = 0 \implies N = 0.$$

Collapse

Normal nilpotent matrices must be zero.

Why Is $r_A(A)$ Normal?

We still owe one property from the trace:

$$A \text{ normal} \implies r_A(A) \text{ normal.}$$

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We still owe one property from the trace:

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This is a polynomial question.

If A commutes with A^H , then powers of A commute with powers of A^H :

$$A^m (A^H)^n = (A^H)^n A^m.$$

So every polynomial in A commutes with its Hermitian conjugate.

Example: Commute a Quadratic Polynomial

Take

$$f(A) = A^2 + 2A + I.$$

Then

$$f(A)^H = (A^H)^2 + 2A^H + I.$$

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$$A^2(A^H)^2 = AAA^HA^H = AA^HAA^H = (A^H)^2A^2.$$

Term by term, the same commuting changes

$$f(A)f(A)^H \quad \text{into} \quad f(A)^Hf(A).$$

Therefore polynomial outputs of a normal matrix are normal.

Normal Implies Diagonalizable

Let

$$N = r_A(A).$$

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The trace now closes:

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collapse	:	$N = 0$

Thus

$$r_A(A) = 0.$$

By the criterion, A is diagonalizable.

Now Spectral Decomposition Is Legitimate

We proved:

$$A \text{ normal} \implies r_A(A) = 0.$$

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Therefore A is diagonalizable.

Now we may use spectral projections:

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k, \quad P_i = f_i(A).$$

Next question: are these projections oblique or orthogonal?

Spectral Projections of Normal Matrices

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Now A is diagonalizable, so previous lectures give:

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Each P_i is the spectral projection onto the λ_i -eigenspace:

$$\text{Im}(P_i) = E_{\lambda_i}.$$

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Each P_i is the spectral projection onto the λ_i -eigenspace:

$$\text{Im}(P_i) = E_{\lambda_i}.$$

And each projection is a polynomial in A :

$$P_i = f_i(A).$$

This is the route: study the projections, not individual eigenvectors.

What the Projections Do

The projections split every vector into eigenspace pieces:

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_k\mathbf{v}.$$

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They satisfy:

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The question is:

Are these projections oblique, or orthogonal?

Normal matrices force the orthogonal case.

What Does a Hermitian Projection Mean?

A projection always splits space into two pieces:

$$\mathbf{v} = P\mathbf{v} + (I - P)\mathbf{v}.$$

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A projection always splits space into two pieces:

$$\mathbf{v} = P\mathbf{v} + (I - P)\mathbf{v}.$$

If also $P^H = P$, then the two pieces are perpendicular:

$$\langle P\mathbf{v}, (I - P)\mathbf{w} \rangle_H = (P\mathbf{v})^H (I - P)\mathbf{w} = \mathbf{v}^H P(I - P)\mathbf{w} = 0.$$

Hermitian projection = projection onto a perpendicular component.

Polynomial Normality Applies Again

We already proved:

Proposition

If A is normal, then every polynomial $f(A)$ is also normal.

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If A is normal, then every polynomial $f(A)$ is also normal.

Spectral projections are exactly such polynomials:

$$P_i = f_i(A).$$

So normality passes from A to every P_i .

Apply This to Spectral Projections

Each spectral projection is a polynomial:

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But P_i is also a projection:

$$P_i^2 = P_i.$$

Now we need one lemma: normal projection $\Rightarrow$ Hermitian projection.

Normal Projection Lemma

Lemma

If

$$P^2 = P \quad \text{and} \quad PP^H = P^H P,$$

then

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In words:

A normal projection cannot be oblique.

It must be a Hermitian orthogonal projection.

Proof: Measure the Obliqueness

Let

$$M = P - P^H P.$$

If $M = 0$, then

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If $M = 0$, then

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Taking Hermitian conjugates:

$$P^H = P^H P.$$

So $P = P^H$.

Goal: prove $M = 0$.

Proof: Positivity Kills the Obliqueness

Compute:

$$MM^H = (P - P^H P)(P^H - P^H P).$$

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Expand:

$$MM^H = PP^H - PP^H P - P^H PP^H + P^H PP^H P.$$

Using

$$P^2 = P, \quad (P^H)^2 = P^H, \quad PP^H = P^H P,$$

each term simplifies:

$$MM^H = PP^H - PP^H - PP^H + PP^H = 0.$$

By positivity, $MM^H = 0 \Rightarrow M = 0$.

Therefore Spectral Projections Are Hermitian

For a normal matrix A :

1. Spectral projections are polynomials: $P_i = f_i(A)$.
2. Polynomials of normal matrices are normal.
3. So each P_i is normal.
4. Each P_i is a projection: $P_i^2 = P_i$.
5. Normal projection lemma gives $P_i = P_i^H$.

Conclusion

For normal A , all spectral projections are Hermitian symmetric.

Hermitian Projection Means Orthogonal Projection

This is the same logic as the real case from the projection lecture:

$$P^2 = P, \quad P^T = P \implies \text{Col}(P) \perp \text{Null}(P).$$

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This is the same logic as the real case from the projection lecture:

$$P^2 = P, \quad P^T = P \implies \text{Col}(P) \perp \text{Null}(P).$$

Now we replace transpose by Hermitian conjugate:

$$P^2 = P, \quad P^H = P \implies \text{Col}(P) \perp_H \text{Null}(P).$$

Same theorem, complex geometry.

Why? The Same Proof with H

General fact:

$$\text{Col}(B^H) \perp_H \text{Null}(B).$$

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Indeed, if $\mathbf{y} = B^H \mathbf{u}$ and $B\mathbf{x} = 0$, then

$$\mathbf{y}^H \mathbf{x} = (B^H \mathbf{u})^H \mathbf{x} = \mathbf{u}^H B\mathbf{x} = 0.$$

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Apply this to $B = P$ and use $P^H = P$:

$$\text{Col}(P) = \text{Col}(P^H) \perp_H \text{Null}(P).$$

Immediately: Eigenspaces Are Perpendicular

Take

$$\mathbf{v} \in \text{Im}(P_i), \quad \mathbf{w} \in \text{Im}(P_j), \quad i \neq j.$$

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$$\mathbf{v} \in \text{Im}(P_i), \quad \mathbf{w} \in \text{Im}(P_j), \quad i \neq j.$$

Then $P_i \mathbf{v} = \mathbf{v}$ and $P_j \mathbf{w} = \mathbf{w}$. Hence

$$\langle \mathbf{v}, \mathbf{w} \rangle_H = \langle P_i \mathbf{v}, P_j \mathbf{w} \rangle_H = (P_i \mathbf{v})^H (P_j \mathbf{w}).$$

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$$\langle \mathbf{v}, \mathbf{w} \rangle_H = \langle P_i \mathbf{v}, P_j \mathbf{w} \rangle_H = (P_i \mathbf{v})^H (P_j \mathbf{w}).$$

Because $P_i^H = P_i$ and compatible projections satisfy $P_i P_j = 0$,

$$(P_i \mathbf{v})^H (P_j \mathbf{w}) = \mathbf{v}^H P_i^H P_j \mathbf{w} = \mathbf{v}^H P_i P_j \mathbf{w} = 0.$$

Therefore different eigenspaces are Hermitian perpendicular.

The Converse Direction

Suppose the spectral projections are Hermitian:

$$P_i^H = P_i.$$

Write the spectral decomposition:

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

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Then

$$A^H = \overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k.$$

Because $P_i P_j = 0$ for $i \neq j$, both AA^H and $A^H A$ equal

$$\sum_i |\lambda_i|^2 P_i.$$

The Projection Criterion

We have proved:

$$A \text{ normal} \iff P_i^H = P_i \text{ for every spectral projection } P_i.$$

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This is stronger than just diagonalizable.

It says:

The spectral decomposition is made of orthogonal projections.

Now the geometry becomes visible.

The Main Story Is Bigger

Hermitian symmetric, skew-Hermitian symmetric, and unitary matrices are all special cases of normal matrices.

But the main theorem is **not merely**:

$$\text{normal} \implies \text{diagonalizable.}$$

That sentence loses the geometry.

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Hermitian symmetric, skew-Hermitian symmetric, and unitary matrices are all special cases of normal matrices.

But the main theorem is **not merely**:

$$\text{normal} \implies \text{diagonalizable.}$$

That sentence loses the geometry.

The real statement has **two parts**:

A normal	$\implies$	diagonalizable and $P_i^H = P_i$ for every spectral projection P_i
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The Projection Version

From previous lectures:

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Meaning: every eigenspace projection is a Hermitian orthogonal projection.

The Geometric Version

Each spectral projection extracts one eigenspace component:

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So the whole space decomposes as an **orthogonal direct sum**:

$$\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} E_{\lambda_2} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.$$

Key: every vector splits into orthogonal eigenvector components.

Unitary Diagonalization

How Do We Build the Diagonalizing Matrix?

We now know:

$$\mathbb{C}^n = E_{\lambda_1} \oplus^{\perp_H} E_{\lambda_2} \oplus^{\perp_H} \cdots \oplus^{\perp_H} E_{\lambda_k}.$$

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To diagonalize A , we need an orthonormal basis inside each eigenspace.

But we already have the eigenspace projections:

$$P_i^H = P_i, \quad \text{Col}(P_i) = E_{\lambda_i}.$$

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But we already have the eigenspace projections:

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Question: How do we extract orthonormal basis vectors from P_i ?

Use the Old Tool Algebraically

From the orthogonal projection lectures:

Previous Result

If $P^2 = P$ and $P^H = P$, then diagonal cross-filling of P produces orthonormal basis vectors for $\text{Col}(P)$.

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If $P^2 = P$ and $P^H = P$, then diagonal cross-filling of P produces orthonormal basis vectors for $\text{Col}(P)$.

Write the produced vectors as colored columns:

$$U = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_r \\ | & | & & | \end{pmatrix}.$$

The algebraic output is

$$P = UU^H = \mathbf{u}_1\mathbf{u}_1^H + \mathbf{u}_2\mathbf{u}_2^H + \cdots + \mathbf{u}_r\mathbf{u}_r^H.$$

Why the Reverse Order Is Identity

The old cross-filling tool gives

$P = UU^H$ and the columns of U are linearly independent.

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Why the Reverse Order Is Identity

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Left-cancel U and right-cancel U^H :

$$U^H U = I_r.$$

This is the AB/BA mechanism from the projection lectures.

Read the Reverse Order as a Table

The reversed product is the inner-product table of the columns:

$$U^H U = \begin{pmatrix} \mathbf{u}_1^H \mathbf{u}_1 & \mathbf{u}_1^H \mathbf{u}_2 & \cdots & \mathbf{u}_1^H \mathbf{u}_r \\ \mathbf{u}_2^H \mathbf{u}_1 & \mathbf{u}_2^H \mathbf{u}_2 & \cdots & \mathbf{u}_2^H \mathbf{u}_r \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_r^H \mathbf{u}_1 & \mathbf{u}_r^H \mathbf{u}_2 & \cdots & \mathbf{u}_r^H \mathbf{u}_r \end{pmatrix} = I_r.$$

So cross-filling produces an orthonormal column table.

Apply It to Every Spectral Projection

For each eigenvalue λ_i :

$$U_i = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{u}_{i,1} & \mathbf{u}_{i,2} & \cdots & \mathbf{u}_{i,r_i} \\ | & | & \cdots & | \end{pmatrix}.$$

The old tool gives

$$P_i = U_i U_i^H = \mathbf{u}_{i,1} \mathbf{u}_{i,1}^H + \cdots + \mathbf{u}_{i,r_i} \mathbf{u}_{i,r_i}^H, \quad U_i^H U_i = I_{r_i}.$$

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Also

$$\text{Col}(U_i) = \text{Col}(P_i) = E_{\lambda_i}.$$

Stack Algebraically

Stack the blocks:

$$\Omega = \begin{pmatrix} U_1 & U_2 & \cdots & U_k \end{pmatrix}.$$

Then

$$\Omega\Omega^H = U_1U_1^H + U_2U_2^H + \cdots + U_kU_k^H.$$

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But these are exactly the spectral projections:

$$\Omega\Omega^H = P_1 + P_2 + \cdots + P_k = I.$$

No geometry is needed here: the identity comes from the projection sum.

Reverse the Order Again

We have a square matrix identity:

$$\Omega\Omega^H = I.$$

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$$\underbrace{\Omega}_A \underbrace{\Omega^H}_B = I \implies B = A^{-1}.$$

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Therefore reverse the order:

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Equivalently,

$$\Omega^H\Omega = I.$$

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Equivalently,

$$\Omega^H \Omega = I.$$

Therefore Ω is **unitary**, and

$$\Omega^{-1} = \Omega^H.$$

Unitary Diagonalization

Each column of Ω is an eigenvector of A .

So $A\Omega = \Omega\Lambda$, where

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \end{pmatrix}$$

with repetitions according to eigenspace dimensions

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with repetitions according to eigenspace dimensions

Multiply by Ω^H :

$$\boxed{\Omega^H A \Omega = \Lambda.}$$

Normal matrices are unitarily diagonalizable.

Example Matrix

Consider

$$A = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

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It is real symmetric, so

$$A^H = A^T = A.$$

Therefore A is normal.

The eigenvalues are

$$0, -3, -3.$$

Spectral Projections for the Example

The radical polynomial is

$$t(t+3).$$

The Lagrange spectral projections are:

$$P_0 = \frac{A+3I}{3} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

and

$$P_{-3} = I - P_0 = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Diagonal Cross-Fill P_0

The projection

$$P_0 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

has rank 1.

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Diagonal cross-filling gives:

$$P_0 = \mathbf{u}_0 \mathbf{u}_0^H, \quad \mathbf{u}_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

So $\mathbf{u}_0$ is an orthonormal basis for E_0 .

Diagonal Cross-Fill P_{-3}

For the -3 eigenspace:

$$P_{-3} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

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Diagonal cross-filling gives one possible decomposition:

$$P_{-3} = \mathbf{u}_1 \mathbf{u}_1^H + \mathbf{u}_2 \mathbf{u}_2^H,$$

where

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

Stack the Rank-One Pieces

We have decomposed the two spectral projections:

$$\underbrace{P_0}_{E_0} = \mathbf{u}_0 \mathbf{u}_0^H, \quad \underbrace{P_{-3}}_{E_{-3}} = \mathbf{u}_1 \mathbf{u}_1^H + \mathbf{u}_2 \mathbf{u}_2^H.$$

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Since spectral projections add to the identity:

$$P_0 + P_{-3} = I.$$

Therefore

$$\mathbf{u}_0 \mathbf{u}_0^H + \mathbf{u}_1 \mathbf{u}_1^H + \mathbf{u}_2 \mathbf{u}_2^H = I.$$

This Is $\Omega\Omega^H = I$

Stack the vectors as columns:

$$\Omega = \underbrace{\begin{pmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 \end{pmatrix}}_{\text{colored eigenvectors}}.$$

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Then

$$\underbrace{\begin{pmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 \end{pmatrix}}_{\Omega} \underbrace{\begin{pmatrix} \mathbf{u}_0^H \\ \mathbf{u}_1^H \\ \mathbf{u}_2^H \end{pmatrix}}_{\Omega^H} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_I.$$

So $\Omega\Omega^H = I$.

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Now Ω is a square matrix.

From previous lectures:

Square Matrix Rule

If two square matrices satisfy $AB = I$, then also $BA = I$.

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Apply this to

$$\Omega\Omega^H = I.$$

Reverse the order:

$$\Omega^H\Omega = I.$$

The reversed order is the inner-product table.

$\Omega^H \Omega = I$ — Inner Product Table

Spelled out:

$$\underbrace{\begin{pmatrix} \mathbf{u}_0^H \\ \mathbf{u}_1^H \\ \mathbf{u}_2^H \end{pmatrix}}_{\Omega^H} \underbrace{\begin{pmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 \end{pmatrix}}_{\Omega} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{inner product table}}.$$

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That means

$$\mathbf{u}_a^H \mathbf{u}_b = \delta_{ab}.$$

This proves the stacked eigenvectors are orthonormal.

Assemble the Unitary Matrix

Collect the orthonormal eigenvectors:

$$\Omega = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}.$$

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The columns are orthonormal, so

$$\Omega^H \Omega = I.$$

This is not an arbitrary eigenvector matrix. It is unitary.

The Diagonalization

The columns of Ω are eigenvectors with eigenvalues

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Therefore

$$\Omega^H A \Omega = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}.$$

Diagonal cross-filling of spectral projections produced the unitary diagonalization.

Eigenvalue Geometry

Return to the Special Cases

We defined three special kinds of normal matrices:

Type	Matrix condition
Hermitian symmetric	$A^H = A$
Skew-Hermitian symmetric	$A^H = -A$
Unitary	$A^H A = A A^H = I$

Return to the Special Cases

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Now normality lets us read these conditions from eigenvalues.

The matrix equation becomes a picture in the complex plane.

Start from Normal Spectral Decomposition

For a normal matrix,

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k, \quad P_i^H = P_i.$$

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Taking Hermitian conjugate:

$$A^H = \overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k.$$

So A^H acts on the same orthogonal eigenspaces, but with conjugated eigenvalues.

Key: special matrix equations become restrictions on each λ_i .

Hermitian Symmetric Means Real Eigenvalues

Suppose

$$A^H = A.$$

Compare the two spectral decompositions:

$$\overline{\lambda_1}P_1 + \cdots + \overline{\lambda_k}P_k = \lambda_1P_1 + \cdots + \lambda_kP_k.$$

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Therefore, for every eigenvalue:

$$\overline{\lambda_i} = \lambda_i.$$

Conclusion

Hermitian symmetric normal matrices have eigenvalues on the **real axis**.

Skew-Hermitian Means Imaginary Eigenvalues

Suppose

$$A^H = -A.$$

Compare:

$$\overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k = -\lambda_1 P_1 - \cdots - \lambda_k P_k.$$

Skew-Hermitian Means Imaginary Eigenvalues

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Compare:

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Therefore, for every eigenvalue:

$$\overline{\lambda_i} = -\lambda_i.$$

Conclusion

Skew-Hermitian normal matrices have eigenvalues on the **imaginary axis**.

Unitary Means Unit-Circle Eigenvalues

Suppose

$$U^H U = I.$$

On the λ_i -eigenspace:

$$U \text{ acts by } \lambda_i, \quad U^H \text{ acts by } \overline{\lambda_i}.$$

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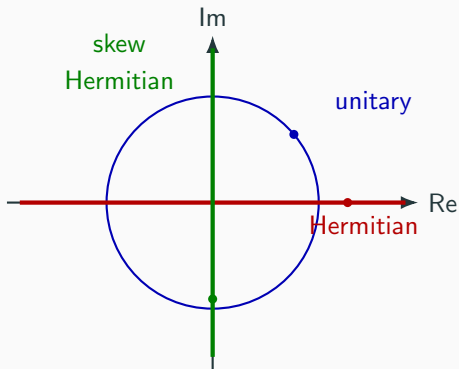
So $U^H U = I$ gives

$$\overline{\lambda_i} \lambda_i = 1.$$

Conclusion

Unitary normal matrices have eigenvalues on the **unit circle**: $|\lambda_i| = 1$.

The Complex-Plane Picture



$A^H = A$	eigenvalues on real axis
$A^H = -A$	eigenvalues on imaginary axis
$U^H U = I$	eigenvalues on unit circle

What Survives Over Real Numbers?

The true machine is complex:

$$A \text{ normal} \implies \Omega^H A \Omega = \Lambda.$$

What Survives Over Real Numbers?

The true machine is complex:

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For real applications, we ask:

When can the whole computation stay real?

That means:

- eigenvalues are real,
- spectral projections have real entries,
- diagonal cross-filling uses real vectors.

Real Symmetric Matrices Stay Real

If A is real symmetric:

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The Lagrange projection formula uses real roots:

$$P_i = f_i(A), \quad f_i(t) = \prod_{j \neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

Since A and the λ_i are real, every P_i is real.

Real Orthogonal Diagonalization

For a real symmetric matrix:

$$P_i^H = P_i \quad \text{becomes} \quad P_i^T = P_i.$$

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Diagonal cross-filling uses only real numbers and produces real orthonormal eigenvectors.

Stack them into a real matrix Q :

$$Q^T Q = I, \quad Q^T A Q = \Lambda.$$

Real Specialization

Real symmetric matrices admit real orthogonal diagonalization.

Why Skew-Symmetric Real Matrices Leave $\mathbb{R}$

If A is real skew-symmetric:

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$$\lambda \in i\mathbb{R}.$$

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Then $A^H = A^T = -A$, so A is skew-Hermitian.

Skew-Hermitian eigenvalues lie on the imaginary axis:

$$\lambda \in i\mathbb{R}.$$

So nonzero eigenvalues are usually not real.

This is why the complex world is necessary.

Real-World Summary

Real matrix type	Complex view	Real outcome
$A^T = A$	Hermitian	real orthogonal diagonalization
$A^T = -A$	skew-Hermitian	imaginary eigenvalues
general real A	may be non-normal	no orthogonal eigenspace guarantee

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Hermitian geometry is the real god: real symmetric theory is its real shadow.

Final Picture of Normal Matrices

Normal matrices are the matrices whose spectral decomposition is orthogonal:

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Special normal matrices add restrictions on eigenvalue locations:

Hermitian	real axis
Skew-Hermitian	imaginary axis
Unitary	unit circle

Key: orthogonal eigenspaces first; eigenvalue geometry second.